

# MEEN 357 Project 1

Jack Knox, Zikama Gedeon, Diego Garza, David Rosales

## 6.1 Coding

Question 1: Yes, there is an advantage to storing the acceleration due to gravity in the planet dictionary rather than hard-coding  $3.72m/s^2$  it inside each function. The project defines `planet['g']` as the gravitational acceleration parameter.

The dictionary simplifies and makes the functions reusable. You could reuse the same rover functions for Earth, the Moon, or another planet by simply changing `planet['g']` rather than editing every function that contains the gravitational constant. It also reduces the likelihood of inconsistent values across functions.

It makes future modifications easier if we keep physical parameters separate from equations.

Question 2:

Calling

`F_gravity(115, rover, planet)`

should produce an **exception/error** rather than calculating a gravitational force. The specifications require `F_gravity` to verify that every terrain angle lies between  $-75^\circ$  and  $+75^\circ$

Therefore,  $115 > 75$ , and the input is outside of the allowed range.

This is desirable behavior because a  $115^\circ$  terrain angle is outside the physical operating range assumed by the rover model. It is possible for the function to return a mathematically valid number that does not correspond to the intended physical situation without validation. Inputs that are rejected prevent misleading results.

## 6.2 Motor and Speed Reducer Behavior

Question 3:

The motor constants are  $\tau S = 170 \text{ N}\cdot\text{m}$ ,  $\omega NL = 3.80 \text{ rad/s}$

From the project specifications

The motor torque-speed relationship is  $\tau = \tau S - \tau S / \omega NL * \omega$

And  $P = \tau \omega$

Therefore,  $P(\omega) = (170 - 170/3.8 * \omega)\omega \rightarrow P(\omega) = 170\omega - 44.7368\omega^2$

For maximum power:  $dP/d\omega = 170 - 89.4737\omega = 0$  giving:  $\omega = 1.90 \text{ rad/s}$

The torque-speed is:  $\tau = 170 - 170/3.8 (1.90) = 85.0 \text{ N}\cdot\text{m}$ .

$P_{max} = \tau \omega = (85)(1.90) \rightarrow P_{max} = 161.5 \text{ W}$

Therefore, an individual rover motor produces 161.5 W, occurring at a motor shaft speed of 1.90 rad/s. The project motor model predicts the parabolic power curve used in this calculation.

Question 4:

Speed reducers increase torque and decrease rotational speed, but they should not change mechanical power.

The gear ratio is:  $N_g = (d_2/d_1)^2 = (0.07/0.04)^2 = 3.0625$

The speed reducer relationships are:  $\tau_{out} = N_g \tau_{in}$  and  $\omega_{out} = \omega_{in} / N_g$

These are specified by the project as input-output relationships.

Therefore,  $P_{out} = \tau_{out} \omega_{out} \rightarrow P_{out} = (N_g \tau_{in})(\omega_{in} / N_g) \rightarrow P_{out} = \tau_{in} \omega_{in} = P_{in}$

At Maximum motor power:

$$\tau_{out} = 3.0625(85) = 260.31 \text{ N} \cdot \text{m} \quad \text{and} \quad \omega_{out} = 1.90/3.0625 = 0.6204 \text{ rad/s}$$

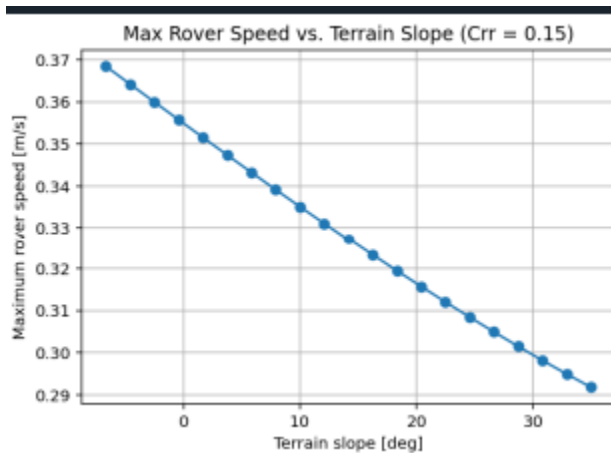
$$\text{Thus: } P_{out} = (260.31)(0.6204) = 161.5 \text{ W}$$

Therefore, the motor and speed-reducer power graphs should show the same maximum power, even though the torque and speed values are different. Models assume an ideal speed reducer and do not include efficiency losses.

### **6.3 Rover Behavior**

#### **Question 5:**

Examine the graph you generated using analysis\_terrain\_slope.py. (Provide the graph in your response or reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how slope impacts rover behavior.



This does make sense physically because as the terrain slope increases the max rover speed decreases. When the slope is greater the power needed to keep at the same speed increases at there is a max power threshold so it makes sense that the max rover speed decreases.

The maximum rover speed falls steadily as slope increases, from about 0.368 m/s at the shallowest downhill angle that has a terminal speed ( $\approx -7^\circ$ ) to about 0.292 m/s at  $35^\circ$ . The trend is nearly linear with very slight curvature, and it makes physical sense.

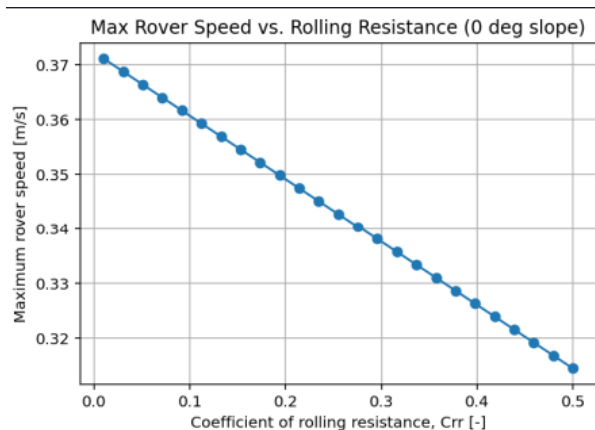
The rover reaches its top speed when it stops accelerating, so the net force is zero: In the model, the motor torque falls linearly with speed, so the total drive force does too

The retarding force is constant for a given terrain, because  $\text{erf}(40v) \approx 1$  at these speeds. As slope increases, the gravity component grows, so the drive force must drop to match it, which happens at a lower motor speed.

The plot has no points below about  $-7^\circ$  because the rover has no terminal speed there. When the rover has no terminal speed, gravity pulls the rover forward harder than rolling resistance holds it back, even at the no-load speed where the motor torque is zero. The rover would keep accelerating, so the code returns NaN and the plot leaves those points blank.

### Question 6:

Examine the graph you generated using `analysis_rolling_resistance.py`. (Provide the graph in your response for reference.) Explain the trend you observe. Does it make sense physically? Why or why not? Please be precise. For example, if the graph appears linear or non-linear, can you explain why it should be the way you observed? Refer back to the rover model and how the coefficient of rolling resistance impacts rover behavior.



This graph also makes sense physically because as speed increases there is more contact with the ground so therefore more resistance on the wheels.

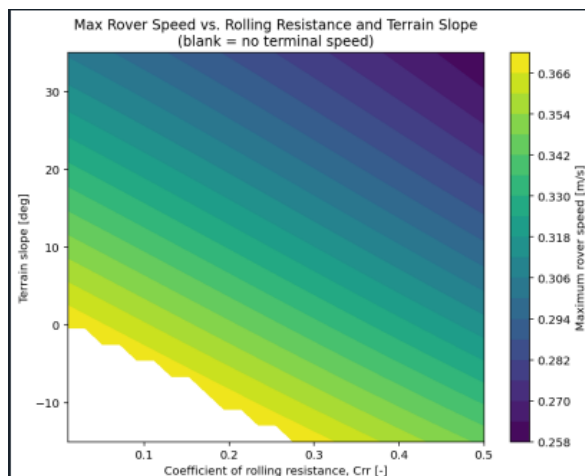
The maximum rover speed decreases linearly as Crr increases, from about 0.371 m/s at Crr = 0.01 to about 0.315 m/s at Crr = 0.5. This makes physical sense, and linearity is expected.

$V_{\max}$  is exactly linear in  $C_{rr}$ , with a slope of about  $-0.116$  (m/s) per unit  $C_{rr}$ . Rolling resistance in this model is  $F_{rr} = C_{rr} (mg) \cos \alpha$ , and it does not depend on speed, since  $\text{erf}(40v) \approx 1$  at these speeds. A larger  $C_{rr}$  (softer ground) gives a larger constant retarding force. The motor's drive force must rise to balance it, and because torque falls linearly with speed, this happens at a lower speed. Both the drive force and the resisting force are linear in the variables involved, so the result is a straight line.

The maximum rover speed decreases linearly as  $C_{rr}$  increases, because elevated rolling resistance generates a greater retarding force, forcing the motor to run at lower rotational speeds to balance the drive force.

### Question 7:

Examine the surface plot you generated using `analysis_combined_terrain.py`. (Provide the graph in your response for reference.) What does this graph tell you about the physical conditions under which it is appropriate to operate the rover? Based on what you observe, which factor, terrain slope or coefficient of rolling resistance, is the dominant consideration in how fast the rover can travel? Please explain your reasoning.



This graph makes physical sense because at the top right corner the maximum speed is the least because there is the most rolling resistance and the most terrain slope, both factors in decreasing maximum speed. The relationship looks linear because the last two graphs both respond linearly to maximum speed.

The surface plot shows that the rover is fastest (about 0.37 m/s) on firm ground (low Crr) and flat or gently downhill terrain. It is slowest (about 0.26 m/s) in the upper right, with steep uphill slopes and soft ground, where both factors add retarding force. The contours are nearly straight, parallel lines, which follows from  $v_{\max}$  being linear in, the same reason the two single-variable plots were linear.

The blank region in the lower left is where the rover has no terminal speed. This is steep downhill terrain with low Crr and where gravity overpowers rolling resistance. Without a speed controller, the rover would accelerate uncontrollably there, so operating in that region is inappropriate. The rover is safe to operate everywhere the surface is filled in, but the entire plot is slow (roughly 0.26 to 0.37 m/s).

The filled-in regions indicate where a terminal-speed solution exists within the model. However, this does not guarantee safe operation because traction, stability, and other physical limitations have not been evaluated.

Slope is the dominant factor over the tested ranges. Moving from  $-15^\circ$  to  $35^\circ$  changes  $v_{\max}$  by about 0.10 m/s, while moving Crr from 0.01 to 0.5 changes it by about 0.06 m/s. The two also trade off directly: since  $d(\sin \alpha)/d\alpha \approx 1$  rad per radian, an increase of 0.1 in Crr costs about as much speed as roughly  $5.7^\circ$  of extra uphill slope. So the slope range tested is simply wider in its effect. Both matter, and soft ground (high Crr) can slow the rover as much as a fairly steep hill.